Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_random_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_163559_d
- started_at_local: 2026-05-13 16:35:59
- finished_at_local: 2026-05-13 16:44:29
- duration_hhmm: 00:09
- duration_seconds: 510.492
- seed_offset: 3000
- replicate_label: d
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.246501	0.040656	0.143578	0.0	0.78	0.0
atsp_random_replay	random	score_only	False	0.190037	0.008454	0.099246	0.841807	0.58	0.013208
atsp_failure_replay	failure	score_only	False	0.195764	0.041579	0.118671	0.0	0.78	0.0
atsp_failure_replay_compression	failure	novelty_gate	True	0.192926	0.009004	0.100965	0.906484	0.58	0.017352

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.246501, synthetic 0.040656, combined 0.143578.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.246501 across 4 instances; family means: ft=0.426937, ftv=0.398016, p=0.015302, ry=0.14575.
- Panel synthetic_holdout mean gap 0.040656 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.077011, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1286, "cost": 1578, "family": "ftv", "name": "ftv33", "optimality_gap": 0.227061}, {"best_known_cost": 1530, "cost": 1863, "family": "ftv", "name": "ftv38", "optimality_gap": 0.217647}, {"best_known_cost": 1473, "cost": 1723, "family": "ftv", "name": "ftv35", "optimality_gap": 0.169722}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.190037, synthetic 0.008454, combined 0.099246.
- Accepted-epoch count 4, mean accepted code novelty 0.841807, and final complexity 0.58.
- Adaptation efficiency 0.013208 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.190037 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004982, ry=0.097143.
- Panel synthetic_holdout mean gap 0.008454 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.195764, synthetic 0.041579, combined 0.118671.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.195764 across 4 instances; family means: ft=0.32281, ftv=0.295102, p=0.019395, ry=0.14575.
- Panel synthetic_holdout mean gap 0.041579 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.080705, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1864, "family": "ftv", "name": "ftv35", "optimality_gap": 0.265445}, {"best_known_cost": 1530, "cost": 1892, "family": "ftv", "name": "ftv38", "optimality_gap": 0.236601}, {"best_known_cost": 1286, "cost": 1511, "family": "ftv", "name": "ftv33", "optimality_gap": 0.174961}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.192926, synthetic 0.009004, combined 0.100965.
- Accepted-epoch count 3, mean accepted code novelty 0.906484, and final complexity 0.58.
- Adaptation efficiency 0.017352 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.192926 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.004093, ry=0.110456.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

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Review of Replay-Aware ATSP Benchmark Suite

Main Evidence Focus: Held-out TSPLIB ATSP Gaps & Synthetic Holdout Gaps

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Mean Code Novelty	Mean Complexity	Notes on Novelty & Transfer
atsp_no_replay	none	0.2465	0.0407	0.1436	0.0	0.78	Baseline no replay; highest gaps; zero code novelty
atsp_random_replay	random	0.1900	0.0085	0.0992	0.84	0.58	Best transfer and synthetic gaps; high code novelty; lower complexity
atsp_failure_replay	failure	0.1958	0.0416	0.1187	0.0	0.78	Moderate gaps; zero code novelty; no compression pressure
atsp_failure_replay_compression	failure + compression	0.1929	0.0090	0.1010	0.91	0.58	Similar transfer to random replay; high code novelty; compression pressure applied

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Interpretation

Transfer Metrics (Held-out TSPLIB & Synthetic Gaps)

- **atsp_random_replay** achieves the lowest mean TSPLIB gap (0.1900) and synthetic gap (0.0085), with best mean transfer gap (0.0992), indicating superior generalization/transfer.
- Both **failure replay conditions** improve over no replay in transfer and synthetic gaps but are slightly worse than random replay.
- **No replay** yields substantially higher gaps across all transfer metrics.

Code Novelty vs Transfer Performance

- Code novelty is zero for no replay and failure replay (no compression), despite improved transfer in failure replay compared to no replay.
- High code novelty (~0.84 and ~0.91) occurs under random replay and failure replay with compression, coinciding with improved transfer gaps.

- This demonstrates that **transfer improves notably with replay (especially random)** even when code novelty is zero (failure replay without compression).
- When code novelty rises (random replay and failure + compression), transfer improves further and complexity decreases.
- Hence, **increased code novelty accompanies transfer improvements but is not necessary for transfer gains**.

Replay Mode and Complexity

- Replay presence (random or failure) reduces final complexity (~0.58) relative to no replay (~0.78).
- Compression pressure is only present in the failure replay with compression condition and coincides with high code novelty and transfer improvements similar to random replay.
- No replay conditions show clustered constructors; replay conditions yield a more balanced final behavior profile.

Conservative Summary

- **Random replay ("atsp_random_replay") is most effective**, yielding lowest TSPLIB and synthetic gaps and highest transfer performance, while substantially reducing search complexity.
- Failure replay without compression improves transfer versus no replay but does not raise code novelty.
- Compression-aware failure replay produces similar transfer benefits as random replay, with even higher code novelty.
- Improvements in transfer and complexity under replay modes occur independently of code novelty in some cases; lexical/code novelty increases do not alone signify algorithmic invention without corresponding metric gains.
- No replay is the least effective condition, reinforcing that replay (random or failure) confers valuable learning benefits.
- Replay mode is a key factor influencing transfer and complexity, with random replay outperforming targeted failure replay.
- Compression pressures may further enable code novelty increases alongside transfer gains, but transfer improvements can be achieved without them.

Key Recommendations

- Prioritize **random replay** for ATSP transfer benchmark tasks to maximize generalization.
- Recognize that **code novelty metrics may not fully explain transfer improvements**; focus on deterministic gaps and complexity measures.
- Carefully distinguish replay modes; replay type impacts transfer and code evolution differently.